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Cyclic Miscible Stimulation Doubles Estimated Ultimate Recovery from Mature Shale Wells

Susan Starr and David Szabo, Energy Transformed, LLC, Houston, Texas, USA; Joseph Dozzo, International Reservoir Technologies, Lakewood, Colorado, USA; Wayne Beecroft BP, Houston, Texas, USA

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Abstract

Application of Enhanced Oil Recovery methods in shale has met limited commercial success to date. The target for improved recovery is huge with typical primary recovery factors in the range of 5% - 20% for most of the shale plays in the US. Cyclic Miscible Stimulation (CMS) is a "Huff and Puff" miscible process which overcomes the two critical challenges faced by Operators through dramatically reduced surface facility costs and dramatically reduced solvent requirements using NGL which is a much more effective solvent than alternative methods. CMS applies Natural Gas Liquids (NGL) in a cyclic process to mobilize residual oil in shale wells. NGL is a superior solvent to other solvents employed to date in the industry because of their lower minimum miscibility pressure. NGL is injected into wells as a liquid, allowing the use of simple low pressure, low rate surface pumping equipment compared to very high pressure, high rate gas injection compressors required for cyclic gas Huff and Puff. Model results backed up by a series of field trials demonstrate that the reservoir physics are robust. Injected NGL effectively mobilizes oil in mature shale wells. History matched models indicate that cyclic injection of NGL over a number of years will result in a doubling or more of EUR for most shale reservoirs.

NGL is a valuable commodity, so the CMS process employs a novel surface facility process flow to capture reproduced NGL and recycle it to make up the majority of the injectant used over the life of a CMS project. The surface facility used for processing produced fluids from the project is standard oilfield equipment configured in a novel process flow that allows the recapture of the majority of the solvent produced from the wells on the "Puff" or production cycle, returning it to the reservoir on the "Huff" or injection cycle. The combination of higher reservoir efficiency with NGL, low-cost facilities, and recycle of solvent allows CMS to deliver better economics than processes using alternative injectants and makes CMS projects competitive with new wells for scarce capital.

CMS has the potential to more than double EUR from existing shale wells. The application of CMS provides huge upside to Operators facing abandonment of mature wells, a dwindling inventory of drilling locations, shrinking proved reserves, and shrinking economic margins on their leases. CMS boosts energy reserves with a fraction of the carbon footprint of alternative sources, greatly enhances the outlook for US energy independence, and delivers tremendous incremental value from the use of relatively low value NGL.

Capturing the benefits from CMS requires access to proprietary methods and facility designs which are covered under a range of US patents.

Introduction

Shale wells decline quickly and Investors struggle with their returns based on primary production alone. Shale producers continue to focus on drilling and fracturing technology with current mergers and acquisitions activity driven by consolidating "Tier 1" drilling locations. For the 2 trillion bbls of shale resources in the USA, <1% have been produced. More than 1 Trillion dollars have already been invested in 100,000+ shale wells and related infrastructure with average recoveries in the range of 5% to 10% of OOIP.

Cyclic Miscible Stimulation (CMS) is transformational technology for enhanced oil recovery in shale and other tight reservoirs as it leverages existing wells and infrastructure and abundant, relatively low value NGL, CMS is a cyclic EOR process using natural gas liquids (NGLs) as the miscible solvent pumped in a "huff and puff" fashion to mobilize additional oil from depleted wells. A nominal volume of 30,000 bbls is injected into a well over a ten day period followed by a production period of 20 – 30 days. This cycle is repeated for up to 5 years delivering incremental production in the 100,000's bbls / well. The CMS process employs equipment that allows a group of 8 – 12 wells to be treated at one time as illustrated in Figure 1. When applied to a group of wells, a CMS project will compete strongly for capital in an Operator's portfolio by doubling the recovery from existing wells for 1/5th of the capex compared to drilling new wells. Production benefits are immediate and reserves can be booked within the first year.

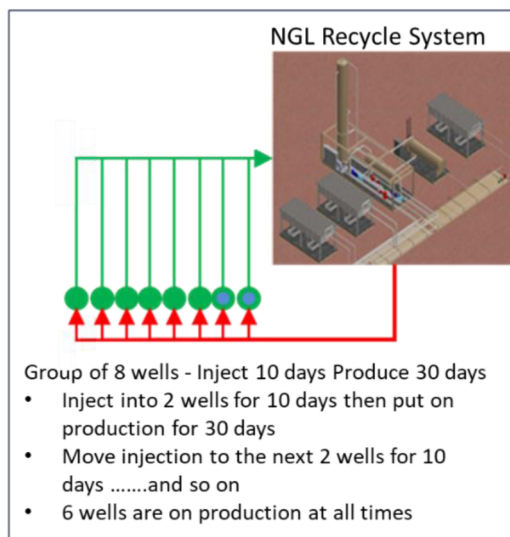


Figure 1—The CMS Process Treats Groups of Wells Injecting for 10 days and Producing for 20 - 30 days per Cycle

Key to CMS success is the ability to recover and reuse the NGLs. The NGL Recycle System (NRS) is modular and mobile and can be moved 4 or more times over its 20 year service life to support multiple groups of wells within an Operator's portfolio. Employing 3 – 4 systems in an Eagle Ford asset with a strong well inventory can deliver a flat production profile of 4,000 bbls / day for up to 20 years. It would take over 100 new wells (\$1 bn) to deliver a commensurate profile.

The Intellectual Property (IP) landscape for executing this technology is fairly complex ranging from the composition of the injected NGLs, the chemistry involved, the modeling techniques, the ability to pump NGLs, and the process of recycling. Nineteen patents have been licensed in an effort to assemble and provide Operators access to the proprietary IP. These patents cover injection of purity and / or unfractionated NGLs for enhanced recovery techniques and the concept of NGL capture for recycling / reuse.

Technology Development

The CMS concept grew out of extensive analysis of a number of Cyclic Field Gas (CFG) EOR projects. Applying CFG as a solvent in "huff and puff" projects in oil shale wells started in the mid-2010s. Huff and Puff EOR projects in shale reservoirs rely almost exclusively on the mobilization of residual oil through miscibility, meaning the injectant (solvent) must be miscible with the residual oil to mobilize oil. These CFG projects proved the reservoir fluid physics and demonstrated the early and longer-term reservoir response and production characteristics. The CFG process continues to be investigated today by many shale oil producers. While some of the projects recovered as much as an incremental 40% over estimated ultimate primary recoveries (EUR) after 8 years of application, the economic results have been mixed. The CFG process requires the use of high pressure compression on surface to achieve and maintain the required downhole pressure of ca 5,000 – 8,000 psi necessary for miscibility, as indicated in Figure 2. Many projects were unable to achieve, let alone maintain pressure above MMP throughout the injection and production cycles, significantly impacting oil recovery efficiency.

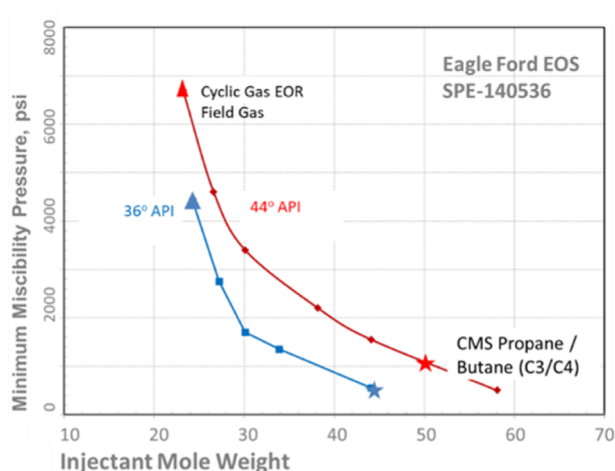


Figure 2—Minimum Miscibility Pressure vs Injectant Mole Weight

Replacing field gas with NGLs as the miscible solvent dramatically lowers the MMP, greatly enhancing the ability of the solvent to mobilize residual oil and increasing the efficiency of the process. In many reservoirs, MMP of the NGLs will be less than current reservoir pressure. This results in first contact miscibility throughout the injection and production cycles, delivering incremental oil volumes of 60% - 150% of primary EUR after cycling for 3 – 5 years. The range of recovery is dependent on the nature of the reservoir fluid. In addition to the reservoir benefits, NGLs are liquid at typical surface processing conditions and are, therefore, pumpable resulting in significantly lower surface facility costs. NGLs are also more easily recovered from the flowback fluids during the production cycle for reuse. Lower facility costs and recycle of injectant are critical factors in project economics.

Fluid physics are the key to the increased efficiency of CMS. When compared to CFG the key variables of oil swelling, viscosity reduction, and density reduction improve with each cycle of NGL injection, as shown in Figure 3. We characterize this phenomenon as "building fluid physics momentum". Reservoir modeling of the CMS process illustrates this concept. The comparison of modeled behavior between CMS and CFG shown in Figure 3 demonstrates why NGLs are much more effective at mobilizing shale oil.

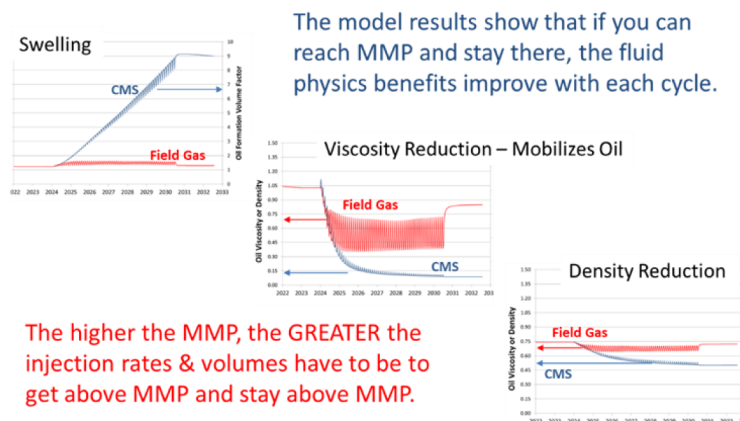


Figure 3—Key Reservoir Fluid Variables for Cyclic NGL (CMS) & Cyclic Field Gas Comparison

A discussion ongoing in the industry is how important the composition of the solvent is to deliver the desired results from a cyclic process such as CMS. Evidence from field trials backed up by reservoir modeling suggests that while constructing a precise composition of solvent through extensive analysis and mixing of components may have academic merit, the practicalities of field operations dictate that the key factor for technical and economic success is having a solvent that is first contact miscible at current reservoir (near wellbore) conditions. It is not necessary to design a solvent made up of refined / fractionated pure components; rather an unfractionated, variable mixture of NGL components, often referred to as Y-Grade, will deliver reservoir benefits while maintaining a low cost of supply and facilitating the ability to recycle components efficiently for reuse.

Inter-well communication, nominally from frac hits, and the impact on system performance is also a topic receiving much attention. Evidence from gas Huff and Puff and CMS field trials indicates fairly rapid pressure communication with nearby offsets with known frac hits. This reservoir response highlights a key advantage of the CMS process over gas HnP. With gas HnP the gas has a propensity to very rapidly move to low pressure sinks, making it virtually impossible for pressure in the project area to be maintained above MMP without making significant accommodations such as establishing a "pressure curtain" around the project area. In the case of CMS, solvent migration, if managed, can be a benefit as there is potentially more surface area being exposed to the solvent which is always above MMP even in depleted areas. Reservoir modeling supports these concepts.

In CMS trials exhibiting pressure communication, the pressure response was followed by increases in production within days or weeks. As modeled, it is assumed that CMS treats the currently connected stimulated rock volume (SRV). History matching field trials indicates that this is a good assumption. As a result, the uplift in recovery from CMS tends to be correlated with primary EUR which is of course correlated with SRV. Better frac designs lead to greater EURs and correspondingly greater response from CMS. We believe the repetitive cyclic injection of NGLs and the resulting propagation of the fluid physics will "re-stimulate" SRV that has closed down over the years, possibly even reconnecting more SRV with offset wells. The greater the SRV, the greater the reservoir surface area exposed to the NGLs. Reservoir simulation suggests wells that are interconnected will need to be managed differently than wells that are isolated. Understanding how wells are responding to each other will be key to optimizing all wells in a given CMS project.

Another frequently raised question is how much of an existing, depleted wellbore will actually see the solvent. One school of thought is that the solvent will exit the injection well at the heel with limited effect along the length of a horizontal well. Another school of thought is that because of the low density and mobility of the solvent, it will preferentially move to lower pressure and it is unlikely that large volumes would simply move into the heel without some distribution along the wellbore or at a minimum through the

established fracture system. There is limited empirical data in the public domain, however there is growing anecdotal evidence from gas HnP projects that a surprisingly large fraction of the wellbore does see solvent on the injection cycle. In any event, there is merit in considering a pre-project wellbore stimulation for each well to be targeted with CMS to ensure that the maximum amount of SRV is exposed to solvent.

Model Development

A modeling methodology has been developed to capture the CMS process in both 2D and 3D compositional dual porosity models. We have built models incorporating older gel hydraulic fracs as well as more modern slick water or hybrid frac treatments. To date, the main focus has been the Eagle Ford and many of the fundamental properties, including the Equation-of-State have been taken from SPE 140536. SPE 140536 is also the source for the compaction and dilation tables incorporated in our models.

For any well modelled, reservoir properties for the model are derived from data available for that well. Initial reservoir fluid compositions and potentially heavy-end component properties are adjusted to match the producing GOR and API gravities. Our grid design is modified to capture the frac design and lateral length. An example is shown in [Figure 4](#).

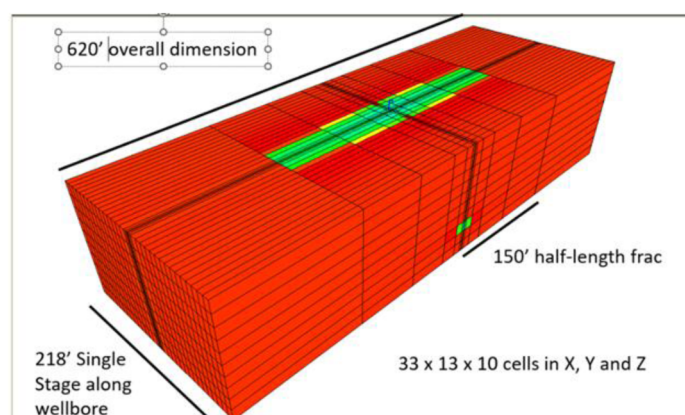


Figure 4—Reservoir Model Grid Design

If a well's production history suggests the well is confined, that is, no evidence of frac hits, we employ a 2D grid and use 3D grids when investigating the impact of frac hits on both historical production as well as predicted CMS recoveries. Our process includes modeling the wells entire production history, including the hydraulic frac treatment(s). We tend to drive model production on total voidage constrained by available pressure data. The history match (or calibration) of any well model is critical to its acceptance as the basis for CMS or other forecasts. We also use the well EUR's as one of our model calibration variables. An example of an Eagle Ford Type well history match is presented in [Figure 5](#).

We recognize no model is unique and use an adaptive history match tool, typically evaluating hundreds, if not thousands of possible combinations of reservoir parameters within confined sets of ranges for properties such as matrix porosity and permeability, hydraulic frac properties such as length and effective conductivity as well as hydraulic frac gridding and impact on numerical dispersion.

The real value in developing these models is to provide guidance on development, implementation, management, and optimization of novel exploitation schemes such as CMS. Of even more value is having a model and methodology that not only matches a conventional production history but also matches reported development schemes such as cyclic lean gas injection and extended to more effective solvents such as NGL's as used in CMS.

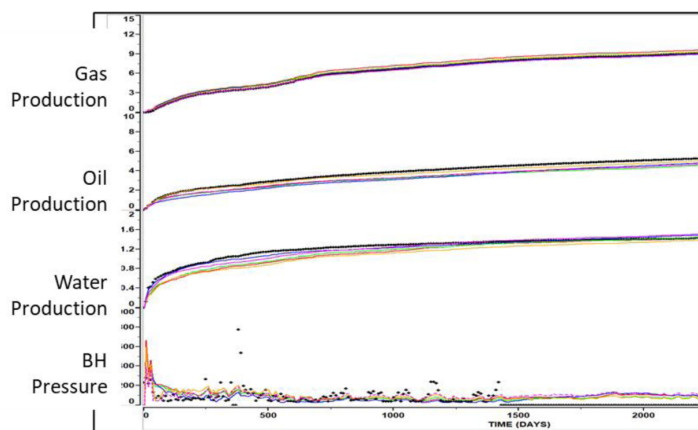


Figure 5—History Matching Gas, Oil and Water Production & Bottom Hole Pressure

Early in our work, we confirmed our model matched the reported results from public domain lean gas injection programs (Cyclic Field Gas (CFG) Huff and Puff projects). Early CFG projects, several of which lasted multiple years, proved up the reservoir physics of cycling a miscible gas in shale reservoirs. They proved miscibility (swelling, viscosity, and density), injectivity, production response, decline and flowback. All key variables that had to be addressed in the model. With this background we used our modeling methodology to analyze a disappointing 18 month CFG multi well trial to explain why it failed to deliver the predicted response. Figure 6 demonstrates the potential for replacing lean gas injection with CMS using the same injection volumes.

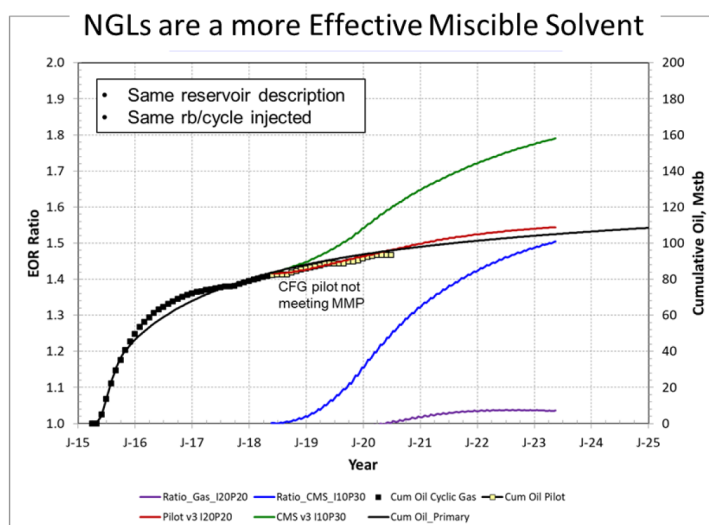


Figure 6—Actual Production Response for an Eagle Ford Cyclic Field Gas HnP Trial and Comparable Predictive Response from CMS

Most recently, our modeling methodology was used to history match primary production, then predict and ultimately match a Cyclic NGL pilot project, SPE-224381-MS. A detailed paper on the modeling of this 2023 NGL / CMS pilot project will be the subject of a future SPE paper. Results of the pilot are discussed in more detail later in this paper under the heading "Field Trials".

Having constructed credible models demonstrated to match real field data, we were able to undertake parametric studies to investigate the importance of operational and reservoir related variables. Some of the variables investigated include: 1) cycle times: injection vs production days, 2) cycle volumes: varying volume of solvent per cycle, 3) injection rates, 4) injectant composition, and 5) reservoir fluid types. None

of these variables are independent however this analysis afforded us the opportunity to focus on which are most important in driving project economics and practical operational issues / constraints.

A summary of insights from the parametric studies are:

- The composition of the injectant is not an important factor as long as the components are NGL components Ethane (C2) and heavier and can be captured in the surface equipment for recycle.
- The CMS process is effective over a wide range of reservoir oil gravity.
 - A good rule of thumb for scoping work is to assume that the CMS process will double ultimate recovery from a well after cycling for 3 – 5 years. Higher API gravities will have a lower uplift while lower gravity APIs will have a higher uplift.
 - A second good rule of thumb is that the amount of NGL "consumed" in the CMS process is approximately 1 bbl for each barrel of incremental oil recovered. Much of that NGL (30%+ is recovered in the blowdown (continued primary production) phase after the cycling project has ceased.
 - Higher gravity crudes produce more propane and butane in the associated reservoir gas, contributing significantly to the volume of solvent injectant available. This reduces the amount of purchased solvent that is required.
- Cycle volume is important, however:
 - The "ideal" cycle volume is not required to be available from day one of the project. Starting with a smaller cycle volume / daily rate allows the project to ramp up over the first few cycles by incorporating recycled solvent. We call this process "Bootstrapping". Employing bootstrapping greatly reduces the logistical challenges of getting solvent to the injection site.
 - Greater solvent volumes will likely result in greater incremental recovery however it will also drive larger equipment handling requirements and costs. There is an economic trade off which will result in an optimization of variables.
- Soak time was investigated. Because the NGLs are first contact miscible and miscible throughout the injection and production cycles, we saw no benefit in soak times. In fact, soak time can be detrimental as it allows the pressure to dissipate which impacts the well's ability to flowback during the production cycle.
- Varying injection rates and pressures was also investigated. There are several SPE papers that attribute benefit with increased injection rates and pressures, specifically URTeC: 2570 and SPE-174946-MS which focus on connecting or reconnecting the induced fractures along with connecting the induced fractures with the natural fracture system. We agree, however, short of including compaction / dilation and hysteresis, which we have done, it is very difficult to model these attributes effectively other than by assuming a reduction in skin damage.

Example graphics from the parametric studies can be found in the Appendix.

Field trials

Field trials started in 2020 and ranged in volumes of several hundred barrels to several thousand barrels of liquid propane & butane mixtures, readily demonstrating the following:

- miscibility of propane & butane at reservoir pressures below 500 psi (Figure 7)
- volatility of production flowback
- effectiveness of proprietary surfactants and paraffin inhibitors
- flowback rates and composition of returned NGLs (Figure 8)

- injectivity of unconventional tight and shale reservoirs (Figure 9)
- ease of pumping NGLs in unconventional reservoirs and staying below frac pressure
- ability to use diversion
- downhole communication with offset wells
- immediate production response

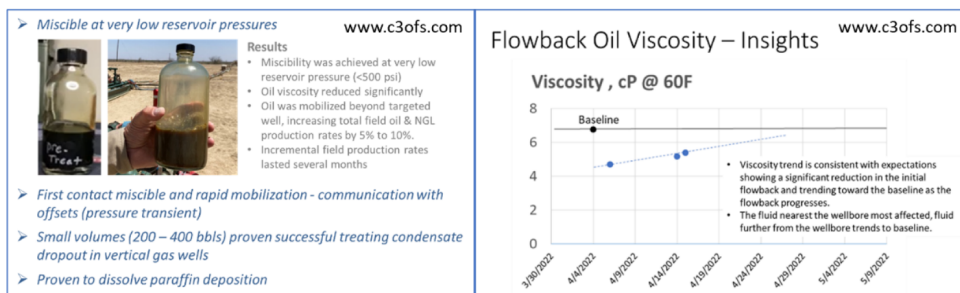


Figure 7—Flowback Samples from Propane/Butane Treatments Demonstrate Miscibility in Depleted Reservoirs

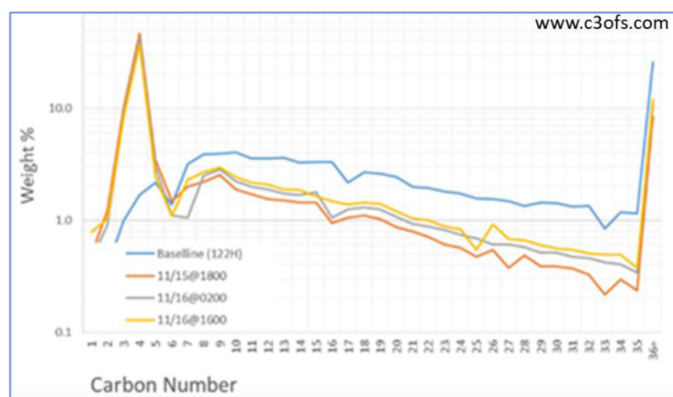


Figure 8—Demonstration of Compositional Analysis of Oil Flowback Samples Changing over Time

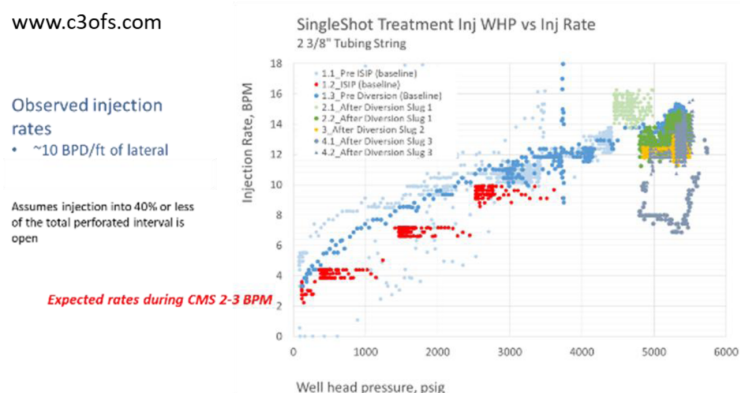


Figure 9—Demonstration of injection Rates and Pressures for a multi 1,000 bbl Treatment in an Eagle Ford Horizontal well with 3 Stages of Diversion

The most comprehensive cyclic field trial using NGL as a solvent occurred in 2023 performed by an Eagle Ford Oil and Gas Company after years of planning (SPE -224381-MS). This single well Eagle Ford horizontal well cyclic NGL trial lasted for 7 months, pumping a total of 170,000 bbls of an unfractionated

Y-Grade NGL in 3 separate high volume cycles averaging 15 days each followed by production periods averaging 45 days each. The trial was designed to test several variables related to project design as well as several operational factors supported by an extensive data gathering program to capture reservoir response and facilitate reservoir model calibration. Production results shown in Figure 10 are the most extensive results to date from a CMS project.

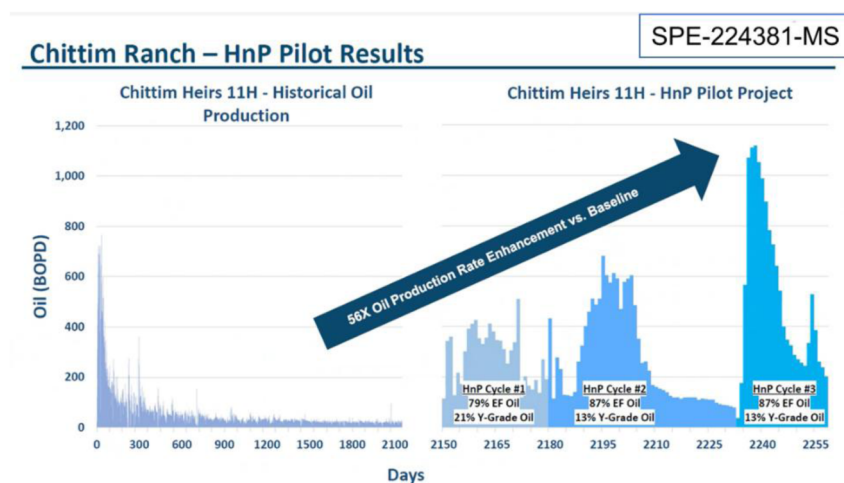


Figure 10—Production Response from a Single Well Multi Cycle Eagle Ford Trial that Lasted 7 Months

The Initial Production (IP) for the pilot well was 760 bopd for one 24 hour period and after 7 years the production rate had declined to 17 bopd prior to the trial. Following the 3rd cycle of NGL injection, the well had a five day period that averaged over 1,000 bopd of incremental production. Cumulative oil production at the start of the trial was 112,000 bbls. The 3 cycle trial delivered an estimated 48,000 bbls of incremental oil. After recalibrating the reservoir model with the extensive amount of technical and compositional data from the pilot, the pre-pilot estimates and post-pilot estimates were within single digit percentage points on incremental oil recovery.

Two main operational challenges handling the returned production from the trial were observed: oil volatility and gas handling / processing volumes, impacting Reid-Vapor-Pressures (RVP) in excess of spec for crude purchasers / haulers. This is because of the high affinity of the propane and butane with the mobilized crude oil. Standard production facilities will struggle to handle the volatility of this production mix.

NGL Recycle Equipment

Critical to the success of CMS is the ability to handle the flowback gas volumes and the high volatility of the crude while recovering NGL (predominantly propane and butane) for reinjection. The NGL Recycle System (NRS) is specifically designed to achieve these objectives. The equipment design specifications are developed from reservoir simulation output interrogating flowback volumes and compositions from the initial, middle and late life segments of project cycles.

The equipment is built on mobile skids and intended to be located as close to the wellhead as possible. It is designed to treat a group of 8 – 12 wells for up to 5 years. After that time, it will be relocated to another group of wells and so on. The expected service life of the equipment is 20 years. The outputs of the NRS are dry gas, stabilized crude for sales, and recovered NGL (predominantly liquid propane and butane) for reinjection. Reservoir simulation predicts the volume of NGLs that are recycled builds up over the first 5 cycles and will average between 65% and 85% of the overall injection demand by the end of the first year of the project. After the first year, the imported solvent demand, or what is termed the "makeup volume", is

expected to be on the order of 20% or less. These modelling predictions are supported by actual flowback data from the 2023 CMS Field Trial.

The NGL injectant composition will be predominantly propane and butane. Y-Grade is the most cost effective NGL, particularly if the NGL is owned by the Operator. If the Operator is using a 3rd party to treat their wet gas, the Operator can negotiate a "take in kind" or "borrow back" agreement to access their own Y-Grade. Once the Y-Grade is introduced into the NGL recycle system, the hexane plus is separated from the propane & butane and is available for oil blending or direct oil sales, creating an immediate commodity sales stream. The propane and butane components are used for solvent in the CMS process. This process reduces the net cost of the solvent to less than \$10/bbl.

Economics

The economics for CMS projects are robust and compete strongly in an Operator's capital investment portfolio. The initial capital investment in a CMS project is associated with the NGL recycling equipment and is comparable to drilling and completing a typical horizontal shale well. This cost is amortized initially over the first 8 wells of the initial project resulting in development costs of <\$5/bbl. Projects mature over 3 – 5 years allowing the surface equipment to be redeployed to new project areas and new groups of wells. Limited additional capex is required when the equipment is relocated and new wells are tied in. Ultimately the total capital investment can be amortized over 30 – 50 wells that will be treated during the equipment's service life. An alternative to purchasing the equipment is to lease the system, minimizing the Operator's initial capex outlay. In this way, the Operator is trading off capex for ongoing opex.

Solvent costs will be greatest during the first 5 cycles of a project. Costs are managed when an Operator can use their own NGLs in the process. This delivers much greater value for the NGLs and reduces the net cost of the solvent. Incremental operating costs are limited to the NGL Recycle System as the system delivers dry gas and stabilized oil to the Operator's production system.

Incremental oil response is instantaneous reaching (and in many cases exceeding) IP30 levels at the beginning of each cycle. Incremental recovery will range from 60% - 150% over primary, depending on the API range of the in-situ oil. Reserves can be booked within the first year. Project payout is generally < 1 year with IRRs typically > 100%. Breakeven oil price is estimated at \$35 / bbl. [Table 1](#) summarizes a comparison of economic parameters for CMS, CFG and Drilling & Completions. Cyclic Miscible Stimulation is very competitive with drilling capex and is more efficient than CFG.

Table 1—Comparison of D&C, CMS and CFG Economic Metrics

	Capex, \$/bbl	Breakeven Oil Price, \$/bbl	Source
Drilling and Completion	8 – 33	46 – 58	Energy News Beat: Aug 4, 2025; www.energynewsbeat.com
CMS	< 5	35	
CFG Historic Projects	15	50+	SPE-200427-MS (2020)

Conclusions

Cyclic Miscible Stimulation (CMS) is transformational technology for enhanced oil recovery in shale and other tight reservoirs as it combines NGL injection with flowback processing and solvent recovery equipment. CMS delivers immediate incremental oil production benefits rivaling initial production rates and it ultimately doubles the well's recovery from existing wells for 1/5th of the cost of drilling new wells.

Cyclic Field Gas trials initiated a decade ago proved up the reservoir physics of cycling miscible gas in shale reservoirs but the economics and containment issues were challenging. Using NGLs in place of lean gas dramatically improves incremental oil recovery as evidenced by the 2023 multi cycle field trial.

NGLs are miscible at pressures well below typical existing shale reservoir pressures so they are first contact miscible throughout the injection and production cycles making them the most efficient solvent choice compared to lean field gas or CO₂. NGLs are pumpable resulting in much lower equipment costs compared to HP compressors. Recycling the NGLs for reinjection improves the economics even more with breakeven oil price of ca \$35/bbl.

Having confidence in the model's ability to forecast CMS response is important. Five years of field trial data collection fed reservoir model development efforts. The model was ultimately tested and calibrated against the 2023 single well multi cycle field trial results, delivering matches in production response, reservoir pressures, and flowback compositions with single digit % calibration values. Parametric studies investigated numerous CMS design variables that have led to optimization of the most impactful variables.

Recovering and reusing the solvent is key to the success of CMS. The NGL Recycle System (NRS) is designed to handle the large volumes of produced fluids from a CMS project of nominally 8 wells as well as "break" the volatility of the crude. One NRS is capable of treating up to 35-50 wells over its 20 year service life, delivering a development cost less than \$5/bbl.

Operators should be mindful of the IP landscape. There are many active patents that protect this proprietary technology as it exists in the CMS arena. When exploring this new technology, Operators need to proactively investigate how they can access the IP in a way that protects them from patent infringement.

Acknowledgements:

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Appendix

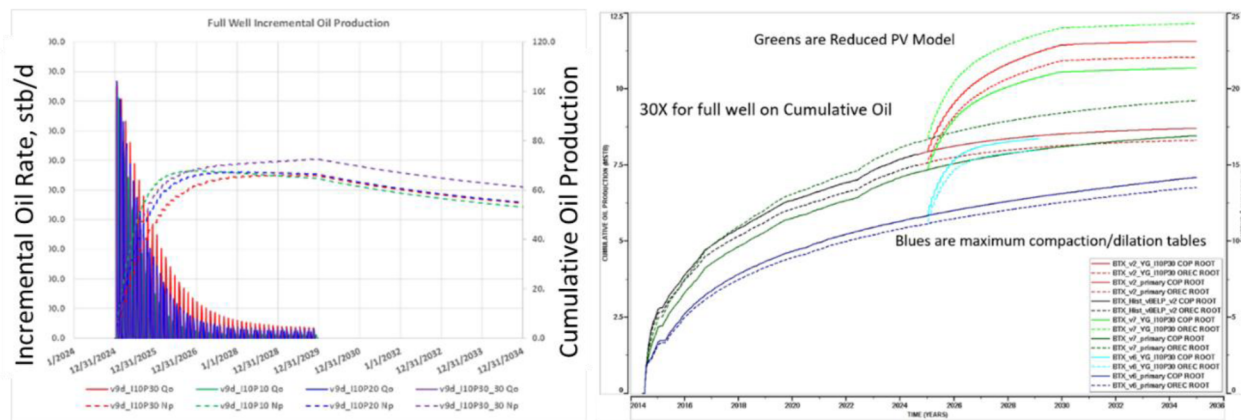


Figure 11—CMS Incremental Production Sensitivities to Injection & Production Cycle Times & Volumes

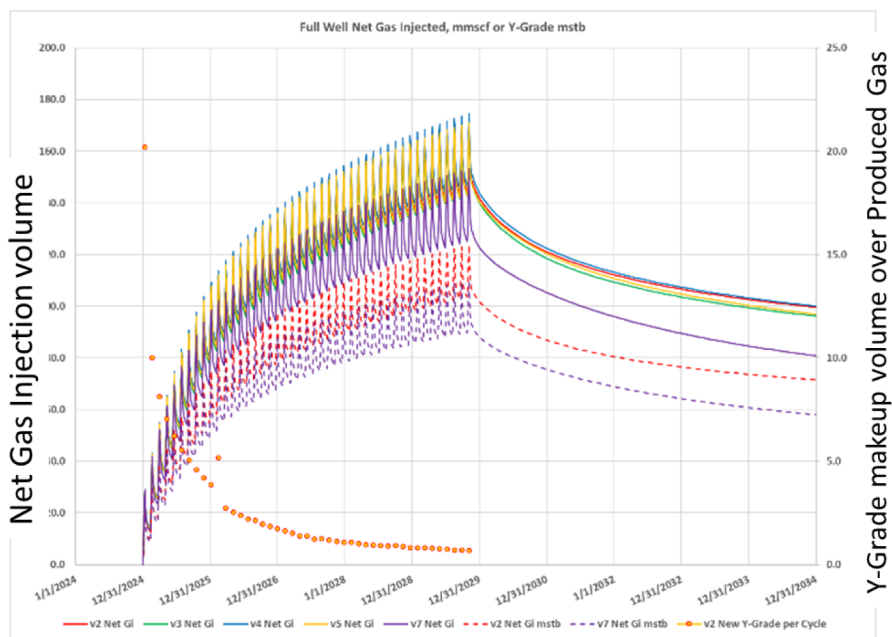


Figure 12—Eagle Ford Volatile Oil Well Response at Varying Injection Volumes